



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

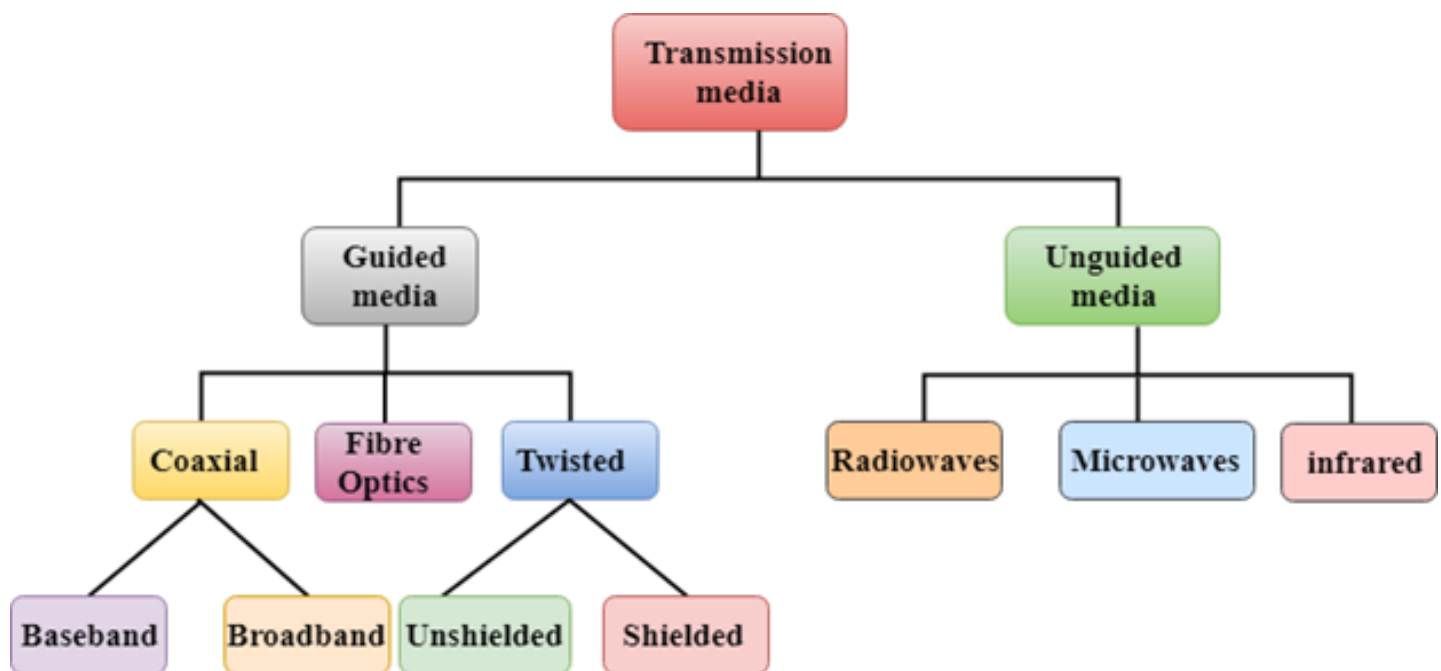
Part -2 (Paper-4)

– Hira Kumar

Computer Networking

Data Communication

Types of Transmission Media



In Computer Network, the sender and receiver are also known by different names.

Name	Other name / different name	
	More use	Less use
Sender	Source / Transmitter	Host / client
Receiver	Destination / receiver	Host / server

2. Unguided Media / Unbounded Media/Wireless

- Unguided media transmits data through **electromagnetic waves** without using a physical conductor.
- This technology is used for data communication because cables are expensive and have high installation and maintenance costs.
- It is also known as wireless communication.
- Signals are normally broadcast through free space and can be received by any device capable of receiving them.
- The transmitter and receiver use antennas to send and receive electromagnetic waves.
- It is easy and faster to install than guided media.
- The signal may be affected by interference, weather conditions, and physical obstacles such as buildings or mountains.
- It is generally less secure than guided media because signals can be **intercepted** more easily.

Intercepted means : the phone call/message is captured,monitored, accessed or listen to by someone/third person in the middle.

Extra Topic (Not in Syllabus)

Q. What is Electromagnetic wave ?

- An electromagnetic wave is a combination of electric and magnetic fields that carries data or energy through free space(vacuum)/air without a physical medium(cable).
- Electromagnetic waves are used in radio, television, mobile networks, Wi-Fi, Bluetooth, satellite communication, wireless Internet, mobile phone calls, GPS, and radar systems.

Types of Electromagnetic waves based on their different frequency ranges.

- i. Radio waves
- ii. Microwaves
- iii. Infrared
- iv. Visible light
- v. Ultraviolet
- vi. X-rays
- vii. Gamma rays

Note : In Computer Networking, the three main electromagnetic waves used for unguided communication are Radio Waves, Microwaves, and Infrared Waves.

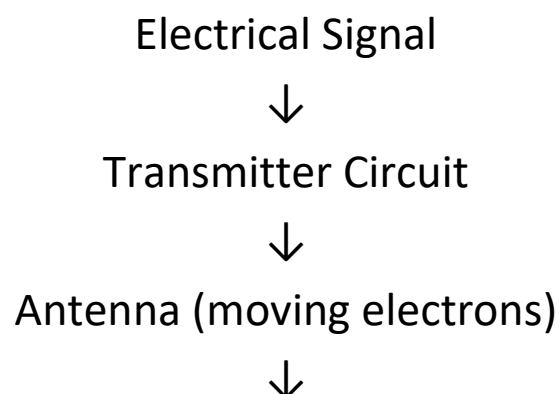
Q. Why are electromagnetic waves used for communication?

Electromagnetic waves can travel through different mediums such as air, water, and vacuum. They do not require any special physical medium for propagation, which makes them suitable for wireless communication.

Q. How Electromagnetic wave works and generate ?

Generation:

1. When **electric current flows through an antenna**, electrons inside the antenna move back and forth (oscillate).
2. Moving charges produce a **changing electric field**.
3. A changing electric field produces a **changing magnetic field**.
4. These changing electric and magnetic fields together form an **electromagnetic wave**.
5. The antenna radiates this electromagnetic wave into free space.



Electric Field + Magnetic Field



Electromagnetic Wave



Receiving Antenna

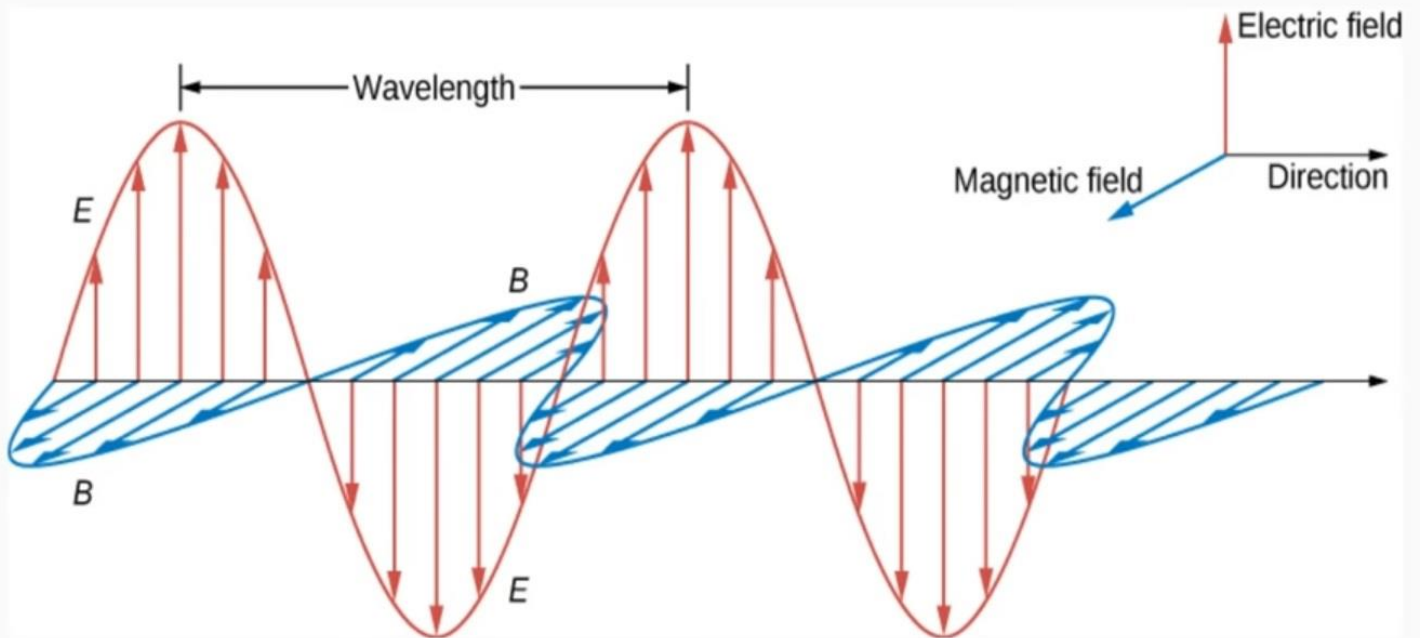


Electrical Signal

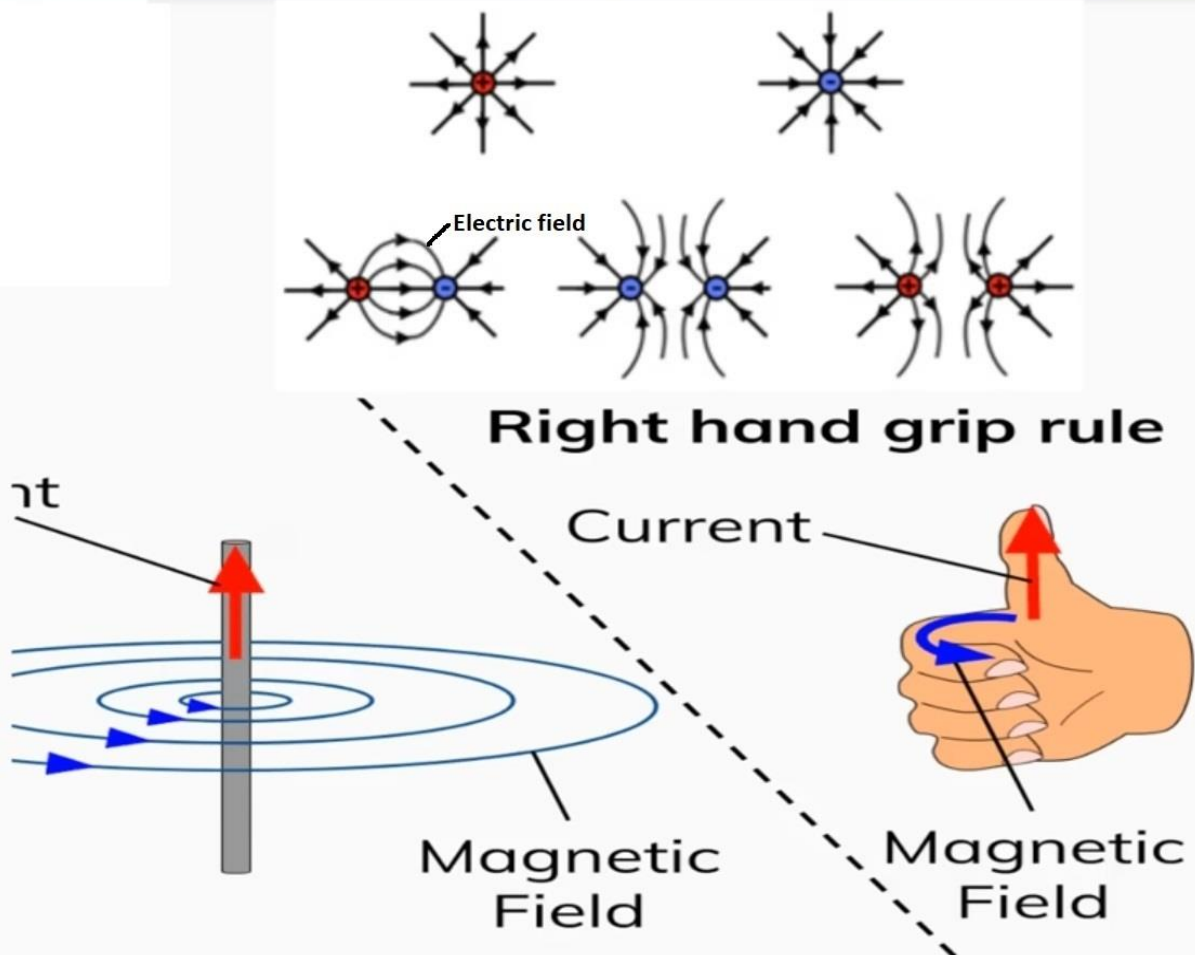
Example (Mobile Phone)

- Your voice → converted into an electrical/digital signal.
- Mobile phone antenna converts this signal into electromagnetic waves.
- Waves travel through air (unguided media).
- The receiver's antenna captures these waves and converts them back into signals.

EM wave



How EM wave Generate



❖ What is frequency ?

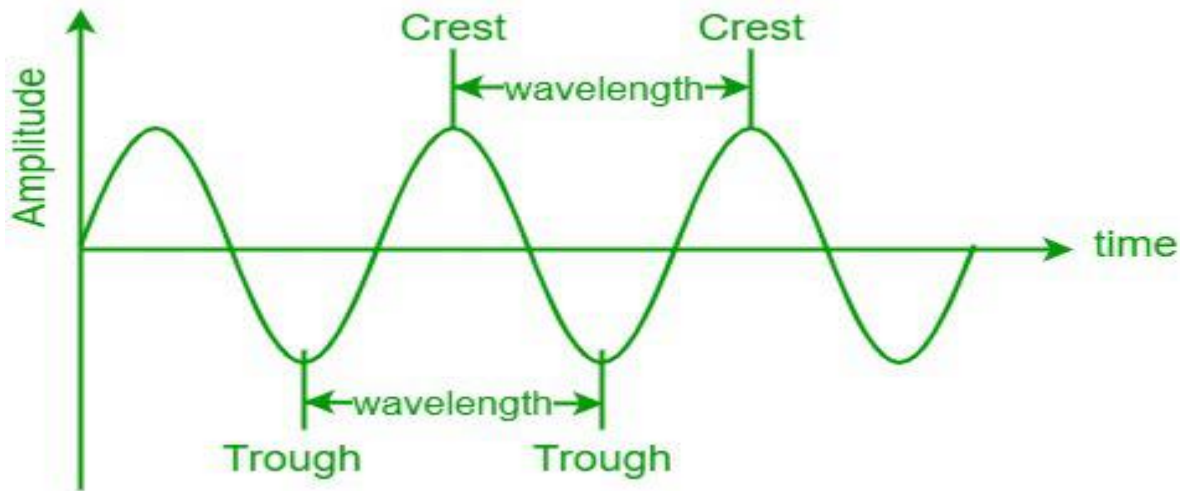
- Frequency is the number of complete cycles or vibrations of a wave that occur in one second.

Or

- Frequency = Number of cycles per second.
- Frequency measure unit is : Hertz (Hz)
 - ✓ 1Hz = 1 cycle per second
 - ✓ 100Hz = 100 cycle per sencond
 - ✓ 1kHz = 1 kilo cycles per second (1000 cycles)
 - ✓ 100KHz = 100 kilo cycles per second (1 lakh)
 - ✓ 1MHz = 1 million cycles per second (10 lakh)
 - ✓ 1GHz = 1 billion cycles per second (100 crore)

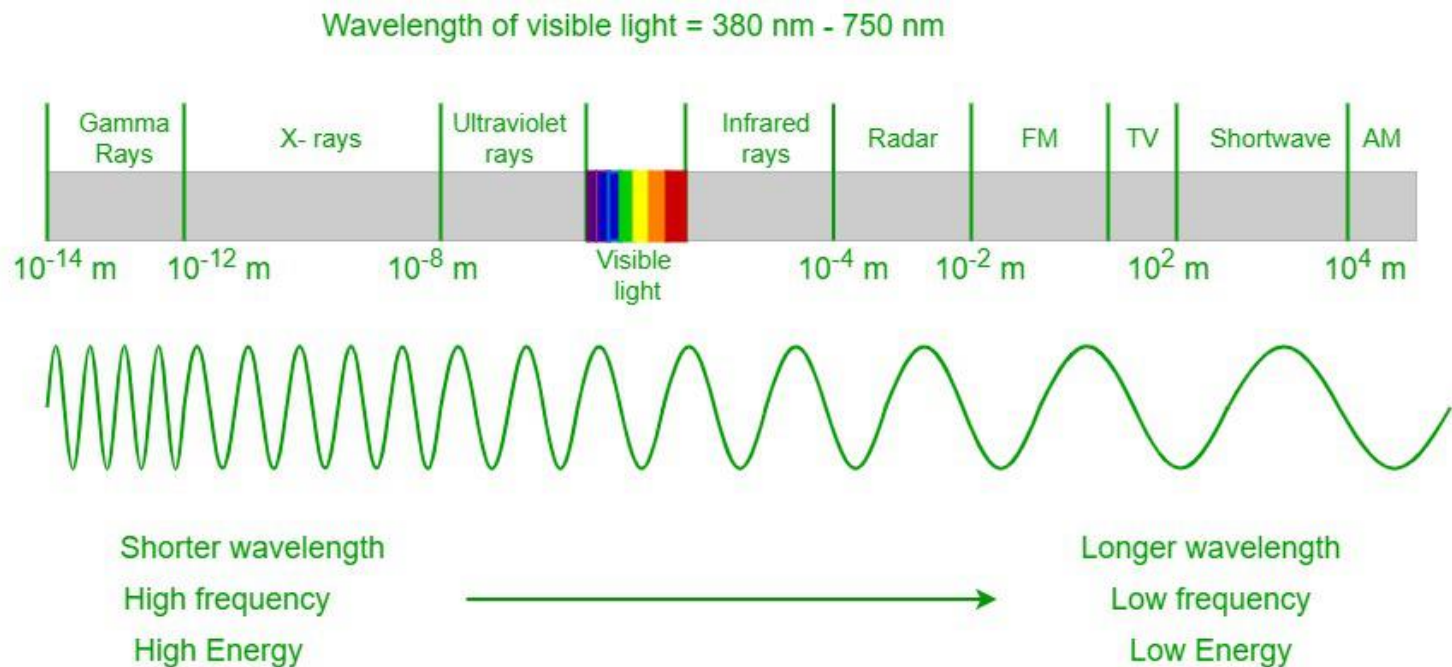
❖ What is wavelength ?

Wavelength is the distance b/w two consecutive identical points on a wave, such as from one crest (top point) to the next crest or one trough (bottom point) to the next trough.

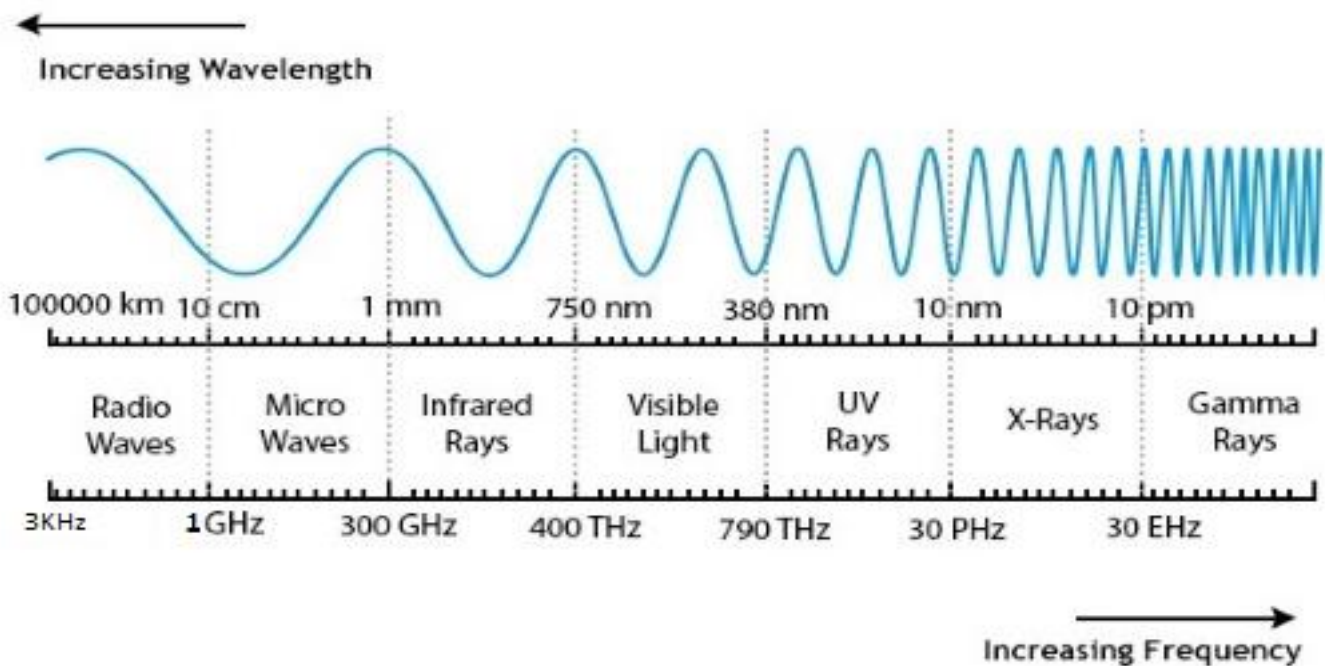


- It is represent by (λ)
- Unit : metre

Wavelength	Frequency	Energy
Increase	Decrease	Decrease
Decrease	Increase	Increase



Electromagnetic Spectrum



❖ What is Electromagnetic spectrum ?

A electromagnetic spectrum is the complete range of electromagnetic waves arranged according to their frequency or wavelength.

Range of Electromagnetic wave = Electromagnetic spectrum.

❖ What is band ?

A band is a specific range of frequencies.

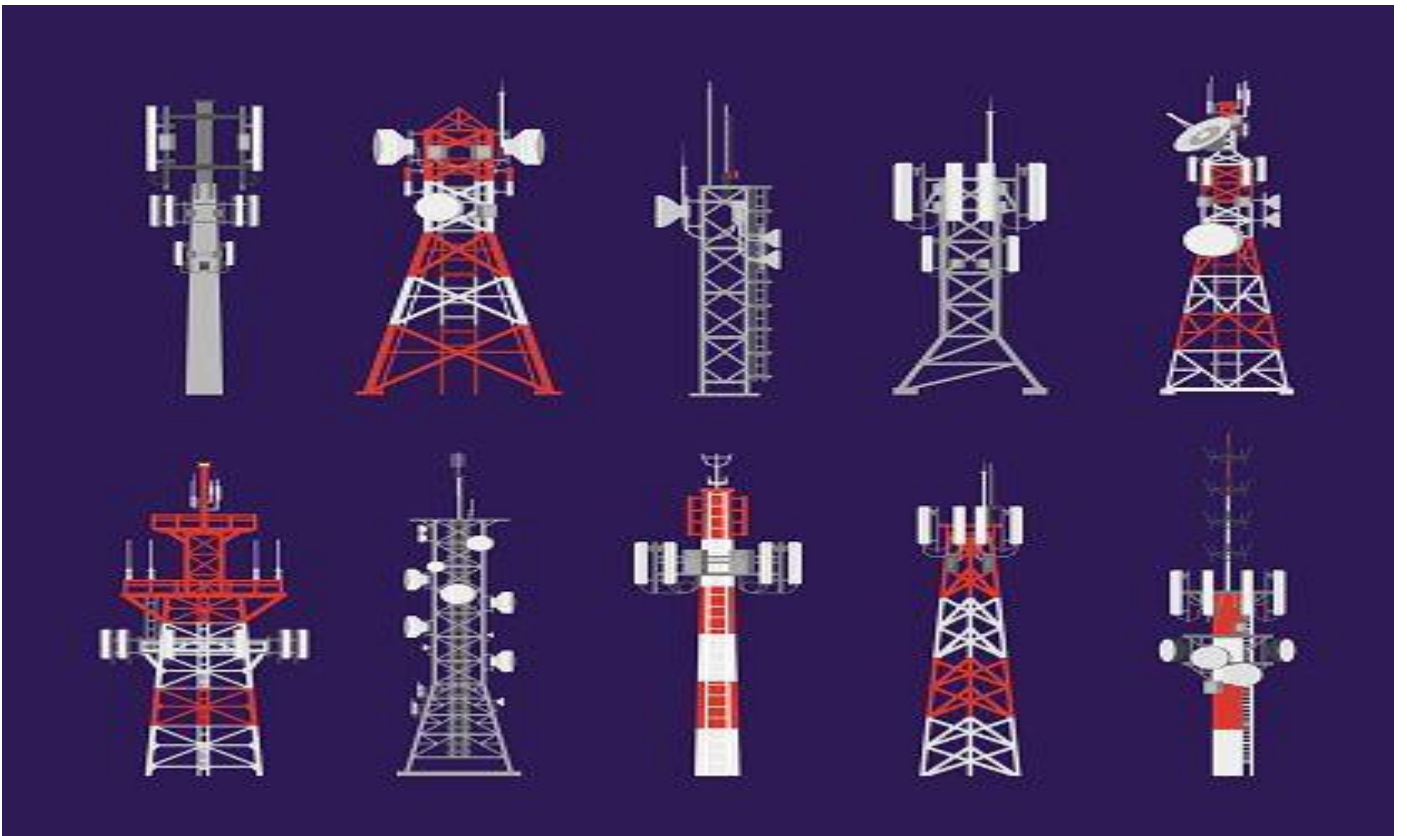
Example : 800-900MHz is a frequency range, which can be called a frequency band.

❖ Antennas

In Unguided media, data is send and receive through antennas. Without antennas not possible wireless communication.

➤ What is antennas ?

An antenna is a device that converts **electrical signals** into **electromagnetic waves** for transmission and converts received electromagnetic waves back into electrical signals.



TYPES of TELECOM ANTENNAS

10 Main Antennas Used in Telecom Networks



Sector Antenna



Omni Antenna



Microwave Dish



Massive MIMO Antenna



Panel Antenna



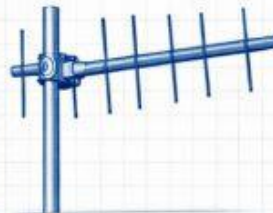
Dish Antenna



Small Cell Antenna



Indoor Antenna



Yagi Antenna



GPS Antenna

❖ **Telecom tower** पर कई प्रकार के **Antennas** लगे होते हैं। हर **antenna** का काम अलग हो सकता है। मुख्य **antennas** ये हैं:

Antenna का नाम और उनके काम (Use)

1. Panel Antenna / Sector Antenna (पैनल एंटीना)

Use : Mobile users (2G/3G/4G/5G) को signal देना और उनसे signal लेना। यह tower के चारों तरफ अलग-अलग sector में coverage देता है।

2. Remote Radio Unit (RRU) + Antenna (आरआरयू)

Use : Mobile signal को process करके antenna तक भेजना और received signal को network equipment तक पहुँचाना।

3. Microwave Dish Antenna (माइक्रोवेव डिश एंटीना)

Use : एक tower से दूसरे tower तक high-speed data link बनाना (backhaul)।

4.Parabolic Dish Antenna (पैराबोलिक डिश)

Use : Satellite communication या long-distance microwave link के लिए

5.Omni-directional Antenna (ओमनी एंटीना)

Use : चारों दिशाओं (360°) में signal फैलाना

6.GPS Antenna (जीपीएस एंटीना)

Use : Tower की location और accurate timing के लिए GPS satellite signal लेना

7.Small Cell Antenna (स्मॉल सेल एंटीना)

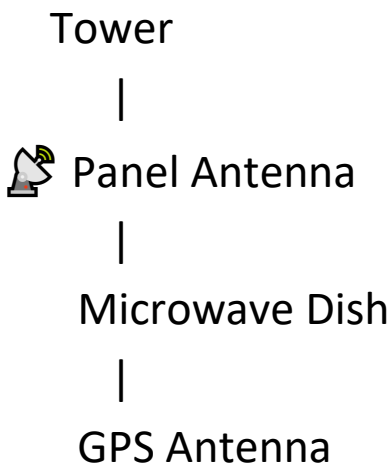
Use : छोटे area (market, stadium, city area) में mobile coverage बढ़ाना

8. **Panel antenna** → आपके mobile से बात करता है

9. **Microwave dish** → tower को दूसरे tower/network से जोड़ता है

10. **GPS antenna** → tower की timing और location maintain करता है

एक सामान्य Mobile Tower पर अक्सर:



❖ What is Wave Propagation (प्रसारण) ?

The way or method by which a signal or electromagnetic wave travels from the sender to the receiver.

Three Methods/Mode are used wave propagation

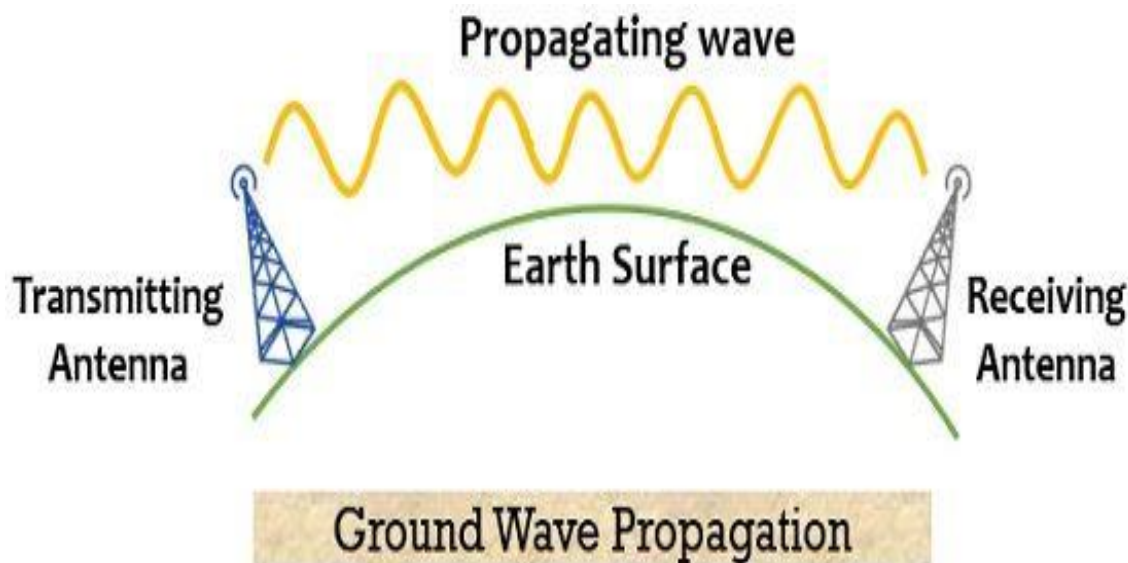
I. Ground propagation

II. Sky-Propagation

III. Line-of-Sight Propagation

I. Ground propagation mode

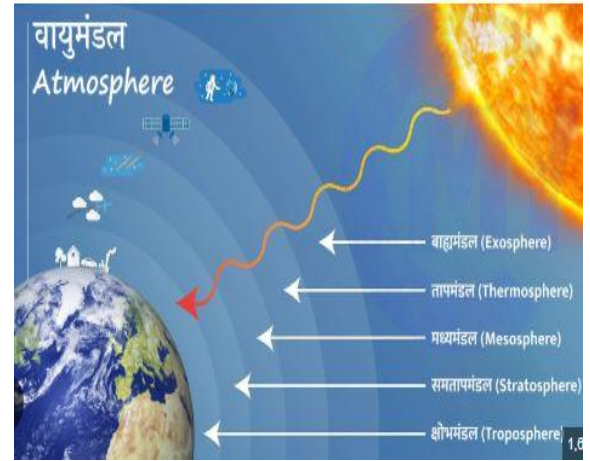
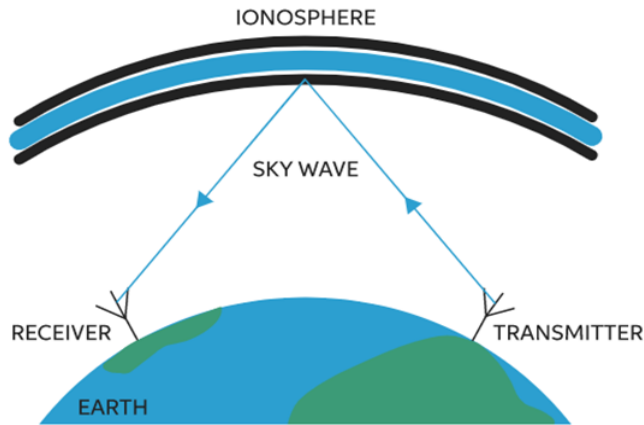
- The Radio waves travel close to the Earth's surface. These low-frequency signals propagate in all directions from the transmitting antenna and follow the curvature of the Earth.
- Example : AM Radio



II. Sky propagation mode:

- In sky propagation, high-frequency radio waves are radiated upward into the ionosphere (a part of the thermosphere), where they are reflected back to the Earth.
- The ionosphere is a layer of the Earth's atmosphere that contains a large number of charged particles (ions).
- This type of propagation allows radio signals to travel over long

distances with relatively low transmission power.



- Example : Shortwave radio (long-distance communication / International radio broadcasting)

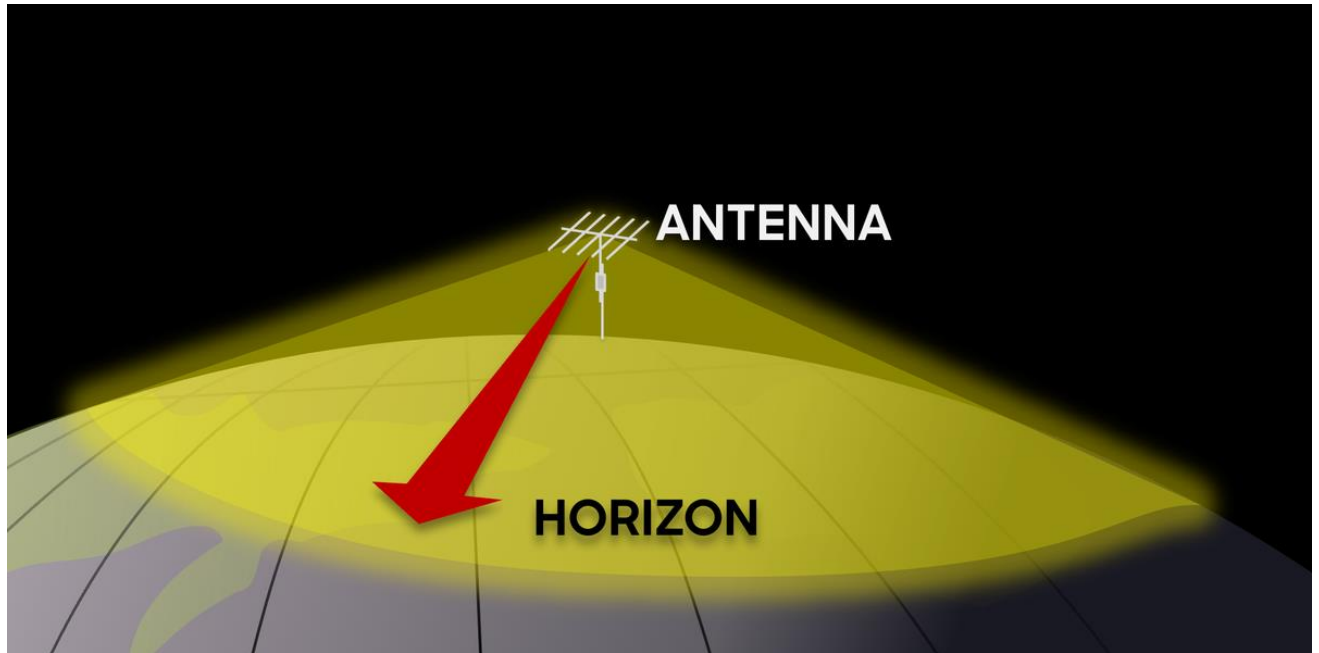


III. Line-of-sight propagation mode / Space wave

- In this mode, very high-frequency radio waves are transmitted in

straight lines directly from one antenna to another.

- The antennas must be directional, face each other, and be tall enough to maintain a clear line of sight..
- Line-of-sight propagation is limited by obstacles and the Earth's curvature because radio waves cannot be completely focused.
- Example : Microwave, mobile towers, wifi, satellite communication



=====HIRA KUMAR=====